

Coordinate Geometry

DSAT Math Practice • 20 Questions • Mixed Difficulty

Total Questions: 20 | Multiple Choice: 16 | Grid-In (SPR): 4

Question 1 [Easy]

A line in the xy -plane passes through the points $(2, 5)$ and $(6, 13)$. What is the slope of this line?

- A) $\frac{1}{2}$
- B) 2
- C) 4
- D) 8

Question 2 [Easy]

In the xy -plane, a line has a slope of 3 and passes through the point $(1, 7)$. Which of the following is the equation of this line?

- A) $y = 3x + 4$
- B) $y = 3x + 7$
- C) $y = 3x + 10$
- D) $y = 3x - 4$

Question 3 [Easy]

The equation of a line in the xy -plane is $4x - 2y = 10$. What is the y -intercept of this line?

- A) $(0, -5)$
- B) $(0, 5)$
- C) $(0, 10)$
- D) $(0, -10)$

Question 4 [Easy]

A line in the xy -plane passes through the origin and the point $(8, -6)$. What is the slope of this line?

- A) $-\frac{4}{3}$
 - B) $-\frac{3}{4}$
 - C) $\frac{3}{4}$
 - D) $\frac{4}{3}$
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Question 5 [Medium]**Student-Produced Response**

Line m in the xy -plane passes through $(3, -2)$ and has a slope of 5. What is the value of b when the equation of line m is written in the form $y = mx + b$?

Question 6 [Easy]

In the xy -plane, the equation $3x + 6y = 18$ represents a line. Which of the following points lies on this line?

- A) $(2, 2)$
 - B) $(3, 2)$
 - C) $(4, 2)$
 - D) $(6, 2)$
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Question 7 [Medium]

The midpoint of a line segment in the xy -plane is $(4, 3)$. One endpoint of the segment is $(1, 7)$. What is the other endpoint?

- A) $(7, -1)$
 - B) $(5, 5)$
 - C) $(2.5, 5)$
 - D) $(3, -4)$
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Question 8 [Medium]

In the xy -plane, which of the following is the equation of a line that passes through $(-2, 4)$ and has an undefined slope?

- A) $y = 4$
 - B) $x = -2$
 - C) $y = -2x + 4$
 - D) $x = 4$
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Question 9 [Medium]

A line in the xy -plane has the equation $y = -2x + 9$. At what point does this line cross the x -axis?

- A) $(9/2, 0)$
 - B) $(9, 0)$
 - C) $(-9/2, 0)$
 - D) $(0, 9)$
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Question 10 [Medium]**Student-Produced Response**

The distance between the points $(1, 3)$ and $(7, 11)$ in the xy -plane is d . What is the value of d ?

Question 11 [Medium]

In the xy -plane, a line passes through $(0, 4)$ and $(6, 0)$. What is the area of the triangle formed by this line and the coordinate axes?

- A) 6
 - B) 10
 - C) 12
 - D) 24
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Question 12 [Medium]

Line p has the equation $5x - 3y = 15$. Which of the following gives the slope and y -intercept of line p ?

- A) Slope = $5/3$, y -intercept = $(0, -5)$
 - B) Slope = $-5/3$, y -intercept = $(0, -5)$
 - C) Slope = $5/3$, y -intercept = $(0, 5)$
 - D) Slope = $3/5$, y -intercept = $(0, -5)$
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Question 13 [Medium]

Student-Produced Response

In the xy -plane, a line segment has endpoints $(a, 1)$ and $(5, 9)$. If the midpoint of the segment is $(3, 5)$, what is the value of a ?

Question 14 [Hard]

The graph of the equation $2x - ky = 14$ in the xy -plane is a line with slope $1/3$. What is the value of k ?

- A) 2
 - B) 3
 - C) 6
 - D) 7
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Question 15 [Hard]

A line in the xy -plane passes through $(-3, 8)$ and $(5, -4)$. Which of the following equations represents this line?

- A) $y = -3/2 x + 7/2$
 - B) $y = -3/2 x + 11/2$
 - C) $y = 3/2 x + 7/2$
 - D) $y = 3/2 x + 11/2$
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Question 16 [Hard]

In the xy -plane, a circle has center $(3, -2)$ and passes through $(7, 1)$. What is the length of the radius of this circle?

- A) 3
 - B) 4
 - C) 5
 - D) 7
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Question 17 [Hard]

Student-Produced Response

The line with equation $ax + 3y = 12$ passes through the point $(6, -2)$ in the xy -plane. What is the value of a ?

Question 18 [Hard]

In the xy -plane, a line intersects the y -axis at $(0, 6)$ and the x -axis at $(4, 0)$. A second line is defined by $3x - 2y = c$. If these two lines intersect at the point $(2, 3)$, what is the value of c ?

- A) 0
 - B) 3
 - C) 6
 - D) 12
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Question 19 [Hard]

Points $A(1, 2)$, $B(5, 2)$, and $C(5, 8)$ are vertices of a triangle in the xy -plane. What is the perimeter of triangle ABC ?

- A) $10 + 2\sqrt{13}$
 - B) $10 + \sqrt{52}$
 - C) $10 + 4\sqrt{13}$
 - D) 20
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Question 20 [Hard]

In the xy -plane, the graph of $y = 2x - 3$ intersects the graph of $y = x^2 - 4x + c$ at exactly one point. What is the value of c ?

- A) 6
 - B) 3
 - C) 0
 - D) -6
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